REAL SOURCE DOCUMENT — MIT COURSE CATALOG

Institution: Massachusetts Institute of Technology (MIT)

Program: Computer Science and Engineering (Course 6-3) — Bachelor of Science

Source URL: https://catalog.mit.edu/degree-charts/computer-science-engineering-course-6-3/

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Date Accessed: March 29, 2025

Department: Electrical Engineering and Computer Science (EECS)

Note: Requirements below reflect the 2022 curriculum. Students entering Fall 2022 or later must use 2022 requirements. Students who entered Fall 2021 or earlier may choose between 2017 and 2022 requirements.

DEGREE REQUIREMENTS — COURSE 6-3 (2022 CURRICULUM)

GENERAL INSTITUTE REQUIREMENTS (GIRs)

MIT requires all undergraduates to complete 17 General Institute Requirements (GIRs), which include:

- Science Core: Calculus I & II (18.01, 18.02), Physics I & II (8.01, 8.02), Chemistry (5.111/5.112), Biology (7.01x)

- REST Requirement (Restricted Electives in Science and Technology): 2 subjects; for Course 6-3, satisfied by 6.1910 (Computation Structures) and 6.1200[J] (Mathematics for Computer Science, if taken under joint number 18.062[J])

- Laboratory Requirement (12 units): Satisfied by one of 6.1010 (Fundamentals of Programming), 6.1020, or 6.2000 in the departmental program

- HASS Requirement: 8 subjects in Humanities, Arts, and Social Sciences including 3-subject concentration plus 2 CI-H subjects

- Communication Requirement: Integrated into HASS; students must also complete 2 CI-M (Communication Intensive in Major) subjects from the departmental program

DEPARTMENTAL PROGRAM (Course 6-3 Core)

The following subjects are required for all Course 6-3 students:

Foundation Subjects:

- 6.100A: Introduction to Computer Science Programming in Python (6 units) — No prerequisite; introductory programming in Python

- 6.1200[J]: Mathematics for Computer Science (12 units) — Prereq: Calculus I (GIR); covers discrete mathematics including logic, sets, graph theory, counting, probability, and proof techniques (cross-listed as 18.062[J])

- Select one introductory EECS subject (9–12 units): 6.1010 Introduction to EECS via Robotics, 6.1020 Introduction to EECS via Medical Technology, or 6.1800 Introduction to EECS via Interconnected Embedded Systems

Core CS Subjects (all required):

- 6.1210: Introduction to Algorithms (12 units) — Prereq: 6.100A and (6.1200[J] or (6.120A and (6.3700, 6.3800, 18.05, or 18.600))); covers mathematical modeling of computational problems, common algorithms, algorithmic paradigms, and data structures; emphasizes relationship between algorithms and programming; enrollment may be limited

- 6.1910: Computation Structures (12 units) — Prereq: 6.1010 or 6.004 equivalent; covers digital logic, finite state machines, instruction set architecture, pipelining, memory hierarchy

- 6.1020: Elements of Software Construction (15 units) — Prereq: 6.100A; covers software engineering principles including design, testing, debugging, and maintenance

- 6.1800: Computer Systems Engineering (12 units, CI-M) — Prereq: 6.1210 and 6.1910; covers computer systems including networking, distributed systems, and performance; communication intensive

- 6.1200[J]: Mathematics for Computer Science (12 units) — described above

Header Electives (choose subjects to complete departmental requirements):

- Select one from: 6.1400[J] Computability and Complexity Theory (Prereq: 6.1200[J] and 6.1210) OR 6.1220[J] Design and Analysis of Algorithms (Prereq: 6.1200[J] and 6.1210)

- Select one from: 6.3700 Introduction to Probability OR 6.3800 Introduction to Inference OR 18.05 Introduction to Probability and Statistics OR 18.600 Probability and Random Variables

Advanced Undergraduate Subjects (AUS):

Students must select at least 2 Advanced Undergraduate Subjects (AUS) from the approved EECS AUS list. At least 1 must be from the Independent Inquiry list. These are upper-division courses numbered 6.3xxx through 6.5xxx.

UNITS SUMMARY

- GIR subjects: ~180 units (standardized MIT unit system)

- Departmental program: 162–171 units

- Unrestricted electives: 48–66 units

- Total units beyond GIRs: 180–183 units

Note: MIT uses a units system (not credit hours). 12 units ≈ 3 credit hours in the conventional semester system.

COMMUNICATION REQUIREMENT

Course 6-3 students must complete at least 2 CI-M (Communication Intensive in the Major) subjects. 6.1800 (Computer Systems Engineering) satisfies one CI-M. Students must select another CI-M subject from the departmental list.

RESIDENCY REQUIREMENT

Students must complete at least 1 year of residency at MIT (2 semesters as a registered full-time student).

SWIMMING REQUIREMENT

All MIT students must pass a swimming test or complete a swimming PE class. This is separate from academic requirements.

SOURCE NOTES

This document reflects publicly available information from the MIT Course Catalog (catalog.mit.edu) for Academic Year 2025–2026. The 2022 curriculum replaced the legacy 6.006, 6.004, 6.009, 6.031, 6.033, 6.042 numbering system with the new 6.1xxx numbering. Legacy course numbers are shown parenthetically where relevant for cross-reference.